

F1 Race Prediction: Model Development and Evaluation

Two-Stage XGBRanker and Plackett-Luce Models — Version 2 (Walk-Forward CV)

0. Inherent Issues in the Original Models and Fixes

Before presenting the updated model results, the methodological issues identified in the original models are documented along with the fixes implemented in version 2.

0.1 Data Leakage in Feature Selection (Critical)

Problem: Both models ran BIC backward, AIC stepwise, and VIF selection on the full dataset — including test seasons — before any cross-validation split. This allowed the model to see future race results when deciding which features to keep, inflating all reported metrics.

Fix in v2: In v2, feature selection is performed exclusively on training data. VIF selection runs on the 2022-2024 reference window and is never given access to the 2025 test season.

0.2 Temporally Invalid Cross-Validation (GroupKFold)

Problem: Both originals used sklearn's GroupKFold($n_splits=5$) grouped by RaceID. GroupKFold does not preserve chronological order — it can assign 2025 races to training and 2022 races to testing, allowing the model to learn future patterns when predicting past outcomes.

Fix in v2: Replaced with walk-forward splits: train on all years strictly before the test year. Five folds are used: three rolling single-year windows (train=[2022]→2023, [2023]→2024, [2024]→2025) and two expanding cumulative windows (train=[2022+2023]→2024, [2022+2023+2024]→2025).

0.3 Wrong BIC Sample Size — PL Model Only

Problem: BIC is defined as $k \cdot \ln(n) - 2 \cdot \ln(L)$ where n is the number of independent observations. For a PL ranking model the unit of observation is a race, not a driver-race row. The original code used $n = \text{len}(\text{model_df})$ (~400 rows) instead of $n = \text{number of races}$ (~22–68), inflating the BIC penalty by ~20×.

Fix in v2: Fixed to $n = \text{train_dff}[\text{RaceID}].\text{nunique}()$ — the correct race count.

0.4 Inconsistent Optimizer — PL Model Only

Problem: Feature selection used L-BFGS-B but the CV loop used BFGS. The two optimisers have different convergence properties, so beta estimates from selection and CV were not directly comparable.

Fix in v2: L-BFGS-B is used throughout in v2.

0.5 Magic Number in Score Adjustment — XGBRanker Only

Problem: The final race score applied an arbitrary 2× penalty to DNF probability: $\text{final_score} = \text{raw_score} - 2.0 \times \text{dnf_prob}$. The XGBRanker already learns the appropriate DNF penalty via DNFProbFeature. The manual adjustment double-counted this and was not derived from any optimisation.

Fix in v2: Removed in v2. The raw ranker score is used directly.

0.6 VIF / LOO Results Discarded — XGBRanker Only

Problem: The original code ran VIF selection and LOO analysis, printed results, then reset FEATURES to the original 18 before training. The analysis had no effect on the deployed model.

Fix in v2: In v2 the VIF-selected feature set is tracked explicitly and both the CV folds and the final model are evaluated consistently.

0.7 QualiPercentile Hardcoded Denominator

Problem: $\text{QualiPercentile} = (\text{QualiPosition} - 1) / 19$ assumed exactly 20 starters. Races with DSQ or DNS entries produced values > 1.0 for some drivers.

Fix in v2: Fixed to $(\text{QualiPosition} - 1) / (n_{\text{starters}} - 1)$ where n_{starters} is the actual grid size per race.

0.8 Infinite Oscillation in AIC Stepwise — PL Model

Problem: The bidirectional AIC stepwise had no guard against revisiting feature sets. With a nearly flat log-likelihood surface, floating-point noise caused AIC to decrease by tiny amounts on each cycle, running indefinitely.

Fix in v2: Fixed by switching to forward-only AIC: greedily add features while AIC strictly improves, stop when no addition helps. Terminates in ≤ 18 steps.

1. Background

Previously, we had been modeling F1 race outcomes using the Plackett-Luce (PL) model. This model is a probabilistic ranking framework that directly estimates the likelihood of a full finishing order. While the PL model gave us an initial approach to model race rankings, there were several limitations.

Qualifying position as we know is a categorical and ordinal variable which in turn caused us to create dummy variables leading our total predictor count going from 6 to 25 with the addition of 19 qualifying position dummy variables. **This caused the qualifying position to capture the full prediction power of the model giving us inaccurate results and unstable coefficients, which is called quasi-complete separation.** We combatted this with adding L2 regularization and splitting qualifying positions into 3 bins (groups). This gave us stable coefficients and evaluation metrics such as Spearman of 50% and top-3 accuracy of 57%, which isn't good enough.

Plackett-Luce Limitations

The PL model **assumed a linear utility structure, which inhibited its ability to capture nonlinear interactions and the complex relationships between the predictor variables.** Along with this, the feature set was restricted to a **small number of pre-race variables** which made it difficult to capture the variation and randomness that occurs in an F1 race.

In the Plackett-Luce model, **each driver is assigned a latent utility representing their underlying competitive strength.** This utility is defined as a **linear combination of pre-race predictors and serves as the main determinant of finishing probabilities.** The model uses these utilities to construct a probability distribution over all possible race outcomes, with higher utility values corresponding to a greater likelihood of finishing ahead of other drivers.

This formula is **inherent to the Plackett-Luce framework**, as it provides a mathematically tractable way to model full rankings using a single scalar quantity per driver. However, this **structure requires that all systematic differences between drivers, which includes both performance and reliability, are captured within this single utility.** Any variation not explained by the predictors is treated as random noise.

In addition to this, the PL model relied on several assumptions. One important assumption was the **independence of irrelevant alternatives (IIA).** This implies that the **ordering between two drivers is unaffected by the presence of other drivers.** In F1, this assumption is constantly violated due to factors like strategy, traffic, and safety cars, all introducing dependencies between multiple drivers.

While these assumptions are convenient, it is **restrictive in the context of Formula 1**, where race outcomes arise from multiple distinct processes. In particular, **reliability-related events such as DNFs are not purely random noise, but instead follow identifiable patterns that are influenced by different factors.**

New Model Motivation

To address these limitations, a **two-staged prediction model that separated performance and reliability into two distinct components** is proposed. **In the first part, a model is used to estimate that a driver DNFs. This captures reliability-related uncertainty.** The second stage is a learning to rank model that predicts the finishing order based on performance related features.

Specific model features include:

- Multi-season telemetry (2022-2025)
- Explicit modeling of driver and team reliability

- ML based ranking framework

Here, the **2022-2025 seasons are chosen**. In **Formula 1 every 3-4 years the rules and regulations change**. What this means is that the FIA (Fédération Internationale de l'Automobile), which is the **governing body of Formula 1, comes up with new rule changes in the way the car should be built**. One example of this is the current season (2026 and onwards), the rules have fully changed and power units now are to be a 50-50 power split between Internal combustion engine (ICE) and electrical power. Before it was an 80-20 split between ICE and electrical power respectively. This is the biggest change with a lot of other changes as well, but this is **done so that the performance gap reduces between teams, and to see more competitiveness**. For this reason the seasons **2022-2025 are chosen as this was the "Ground Effect Era"** where aero focus shifted away from upper-body to underbody ground effects. **Using years with different rules would cause our data to be misinterpreted and give us inaccurate predictions.**

2. Model Framework

Part 1: Reliability Modeling (DNF Probability)

A logistic regression model is used to estimate:

$$P(\text{DNF} \mid \text{pre-race features}) = P(\text{DNF} \mid \text{race conditions} + \text{reliability history} + \text{variability})$$

The **logistic regression model estimates the probability that a driver DNFs (does not finish a race) as a function of pre-race predictors**. These predictors include both **historical reliability measures**, such as driver and team DNF rates, and **race-specific variables**, such as weather conditions, lap-time variability, and interaction effects. **Variability captures the stability of a driver's performance, which doesn't directly cause DNFs, but is associated with increased race risk**. The model combines these inputs through a linear function, which is then transformed into a probability using a logistic function.

This formulation allows the model to capture **how DNF risk varies across different race contexts** rather than relying solely on long-run historical averages. It accounts for the fact that **reliability is influenced not only by historical performance but also by race-specific conditions** such as weather and track characteristics. As a result, the **model estimates a context-dependent probability of failure**, reflecting structured sources of variability rather than treating DNFs as purely random events.

Along with this, logistic regression is suited for this because **DNF is a binary outcome** (finish vs. not finish), and the model directly estimates probabilities constrained between 0 and 1. Its coefficients help us understand how each factor contributes to DNF risk.

Part 2: Ranking Model (Race Outcome Prediction)

The ranking model used is **XGBRanker**, XGBoost's implementation of the LambdaMART learning-to-rank framework. Rather than predicting a finishing position directly, XGBRanker **learns pairwise comparisons between drivers within each race** — given two drivers and their feature vectors, which one finishes ahead? With 20 drivers per race this produces 190 pairwise comparisons per race ($20 \times 19 / 2$), and the model learns across all races simultaneously.

Each driver's feature vector contains all 18 engineered predictors **plus the DNF probability estimated in Stage 1 (DNFProbFeature)**. This means reliability is not a post-hoc adjustment — it is fed directly into the ranking model as a feature, allowing XGBRanker to learn how DNF risk interacts with qualifying position, team strength, and driver form when determining finishing order.

XGBRanker is trained to optimize **NDCG (Normalized Discounted Cumulative Gain)**, a ranking metric that rewards correctly ordering drivers at the front of the field more than at the back. This is appropriate for F1 because errors in predicting the top finishers matter more than errors in predicting, say, P15 vs P16. Unlike linear models, XGBRanker captures **nonlinear relationships and feature interactions** without requiring them to be specified explicitly.

Final Prediction

The final predicted finishing order for each race is determined by **ranking drivers by their raw XGBRanker score** — no manual adjustment is applied. The driver with the highest score is predicted to finish first, and so on. Because DNFProbFeature is already an input to the ranker, reliability risk is embedded in the score itself: a driver with high DNF probability will receive a lower ranking score without any external penalty term being needed.

This contrasts with the original v1 approach, which applied a manual penalty of **final_score = ranking_score - (2 × DNF_probability)** after the model had already learned from DNFProbFeature as an input. That

double-counted the reliability signal. In v2 the formula is simply:

final_score = XGBRanker output score

Drivers are then ranked in descending order of this score to produce the predicted finishing order for the race.

3. Data and Preprocessing

We utilized FastF1 data for the 2022-2025 seasons. Each observation corresponds to one driver × one race.

3.1 Matrix

Unlike the previous Plackett-Luce model, where each race was treated as one ranked observation, the **new learning-to-rank model is built at the driver-race level**. This means each row represents one driver in one race, so the **sample size is the total number of driver-race observations, not just the number of races**. For example, one race with 20 drivers contributes about 20 rows. Across multiple seasons, this creates a much larger feature matrix while still preserving the race-level grouping structure.

This theoretically with 1 race 20 drivers equaling 20 rows. 1 season (~22 races) is 22×20 equaling 440 rows. Now 4 seasons around 90 races × 20 rows (drivers) equaling ~1800 rows. The number of races here around 88 to 92 isn't our n — it is the number of groups. So in our model $n \approx 1800$, groups ≈ 90 , and p (predictor list) $\approx 19-20$.

The model is learning to rank **within** each group. So ranking happens inside of each race (group) but the model learns across all rows.

The feature matrix consists of 19 predictors for the DNF model and adds an additional feature for the ranking model. The additional feature is the predicted DNF probability.

3.2 Predictor Types

A key issue in the PL model was being able to handle qualifying position as a categorical and ordinal variable which created issues.

Qualifying Position

Qualifying position is both categorical and ordinal so:

- **The difference between P1 and P2 is not the same as P10 and P11**
- Treating it as continuous would introduce artificial linear assumptions

To address this, qualifying position was transformed into a percentile representation. In v2 this is computed as:

$$\text{QualiPercentile} = (\text{QualiPosition} - 1) / (n_starters - 1)$$

Note (v2 change): The original formula used a hardcoded denominator of 19, assuming exactly 20 starters. This produced values > 1.0 for drivers on grids with DNS/DSQ entries. The denominator is now (n_starters - 1) where n_starters is the actual grid size per race, clipped to a minimum of 1 to avoid division by zero.

Qualifying position is an ordinal variable, but not linearly spaced. To preserve ordering while avoiding high-dimensional dummy encoding, it is transformed into a normalized percentile. While this transformation introduces a linear scale, the use of a tree-based model allows the relationship between qualifying position and race outcome to remain nonlinear, as the model learns optimal split thresholds rather than assuming constant marginal effects.

Continuous Variables

All continuous predictors are standardized using **z-score normalization**: $x_scaled = (x - \mu) / \sigma$

Each feature is centered by subtracting the mean and divided by its standard deviation. This ensures that all predictors are on a comparable scale. This in turn **improves numerical stability** and enables **reliable coefficient estimation** for stage 1, the **logistic regression model**. XGBoost is a **tree-based model**, meaning

that it **doesn't require scaling**, but standardization is applied for consistency across both stages.

4. Feature Engineering

The model incorporates a comprehensive set of pre-race predictors. All predictors used in the model are either **directly observable prior to the race** (e.g., qualifying results, weather conditions, and practice session telemetry) or **derived from historical race data** (e.g., driver form, team strength, and reliability measures).

4.1 Qualifying Features

- *QualiPercentile* - normalized grid position
- *QualiGapToPole* - continuous measure of performance
- *TeammateQualiGap* - relative intra-team performance

4.2 Telemetry / Practice Features

- *FP_LongRunPace* - proxy for race pace
- *FP_LongRunVar* - consistency indicator

4.3 Performance Indicators

- *DriverForm* - rolling average of recent finishes
- *TeamStrength* - rolling team performance

4.4 Reliability Features

- *DriverDNFRate*
- *TeamDNFRate*

4.5 Environmental Features

- *AirTemp*
- *TrackTemp*
- *Rainfall*

4.6 Track Characteristics

- *OvertakeIndex* - difficulty of overtaking

4.7 Interaction Terms

- *Quali x Overtake*
- *Pace x TireDeg*
- *Reliability x rain*
- *Driver_vs_Team*
- Measures how well driver performs relative to car they have and since standardized, it becomes how many standard deviations above/below team expectation is the driver

These allow the model to capture the nonlinear effects.

5. Model Selection

5.1 Logistic Regression (DNF Model)

Logistic regression is used for the DNF prediction stage. The model gives us a direct estimate of the probability that a driver fails to finish a race. This is essential in correctly integrating reliability into the final prediction. Along with this, logistic regression allows for a clear understanding of how different features contribute to DNF risk.

5.2 XGBoost Ranking Model (LambdaMART)

As mentioned earlier this is a learning to rank framework which is well-suited for Formula 1, given that our goal is to predict the full ordering of drivers over absolute outcomes. Also, XGBoost does well in capturing nonlinear relationships and feature interactions. While other gradient boosting methods like LightGBM or CatBoost could be used, XGBoost already gives us a stable implementation without the need for extensive feature engineering.

As mentioned above, when we do many pairwise comparisons, with 20 drivers we have $C(20,2) = 190$ comparisons. If we used hypothesis testing for each pair (looking at Driver A is better than Driver B), each test has an error rate α (say 0.05). What this means is doing 190 tests then will inflate the chance for a false positive. Formula for at least one false positive: $1 - (1 - \alpha)^g$. Doing 190 tests would then cause the error to be huge, much greater than 0.05 or any alpha. To fix this we use Bonferroni correction to shrink the per-test threshold: $\alpha_{\text{adjusted}} = \alpha / g$.

So instead of comparing each test to 0.05, we compare it to $0.05/190$ or ≈ 0.00026 . What this does is it keeps total error controlled. The difference is in our model, XGBoost and LambdaMART don't actually use hypothesis testing. What this method actually does is optimization, and not p-values, significance thresholds, or α levels. It minimizes a loss function. Pairwise logistic loss: $\log(1 + e^{-(s_A - s_B)})$. So it optimizes metrics such as NDCG. The 190 pairwise comparisons are used as training signals, and not statistical tests. **All pairwise comparisons are jointly optimized.** The model doesn't learn if A is significantly better than B, but instead it learns to adjust scores so correct orderings are more likely overall.

5.3 Training and Evaluation: Walk-Forward Cross-Validation

Why walk-forward CV — and why not standard k-fold?

Standard k-fold cross-validation randomly partitions the dataset into k subsets, trains on k-1 of them, and tests on the remaining one. This works well for data where observations are independent. **F1 race data is not independent across time** — a model trained on 2024 race results has already seen which drivers were fast, which teams improved, and how the competitive order evolved. If we then ask it to predict 2022 races as a "test set," we are evaluating it on data from the past using knowledge from the future. This is **temporal data leakage** — the model appears to perform well not because it learned generalizable patterns, but because it was inadvertently given information it would never have in a real deployment scenario.

Walk-forward cross-validation solves this by enforcing a strict rule: **the test year must always be strictly later than every year in the training set.** The model can only be evaluated on seasons that, at the time of training, had not yet happened. This mirrors real-world use: when predicting a 2025 race, only pre-2025 data is available.

How each fold works

For each fold, the procedure is identical:

1. **Fit the model** on all races in the training years. For XGBRanker this means training the boosted ranking model; for the PL model this means maximizing the penalized log-likelihood over the training races.
2. **Predict** race outcomes for every race in the test year, using only the pre-race features available before that race. The trained model has never seen any result from the test year.

3. **Evaluate** the predictions for that test year by computing Spearman, Kendall, Top-3/5 accuracy, Winner accuracy, and NDCG against the actual race results.

This process is repeated for each of the five folds. The final reported metrics for each model are the **mean of the per-fold scores across all five folds**. Taking the mean across folds is important because any single fold can be unrepresentative — one season may be unusually predictable (Red Bull dominance) or unusually chaotic (2025 competitive reset). Averaging across five folds covering three different test years gives a stable estimate of how the model performs across a range of racing conditions.

The five folds

Two types of fold are used. **Rolling folds** train on a single season and test on the next, measuring how well very recent patterns transfer. **Expanding folds** accumulate all prior seasons as training data, measuring whether more historical data helps. Together, the five folds cover every possible test season (2023, 2024, 2025) and multiple training configurations:

Fold	Train Years	Test Year	Type
1	2022	2023	Rolling
2	2023	2024	Rolling
3	2024	2025	Rolling
4	2022 + 2023	2024	Expanding
5	2022 + 2023 + 2024	2025	Expanding

Table 5a. Walk-forward cross-validation splits.

The 2025 season appears in both Fold 3 (rolling, trained on 2024 only) and Fold 5 (expanding, trained on 2022-2024), allowing a direct comparison of whether a single recent season or three seasons of history gives better 2025 predictions. At no point does any fold use 2025 data for training — it is always held out as the final out-of-sample test period.

5.4 Plackett-Luce Models

The Plackett-Luce model is included as a baseline probabilistic ranking model. Three iterations of the PL model were developed before the current v2 framework:

5.4.1 Original PL (2024 season only)

This was the original Plackett-Luce model that was created. This model features such as *qualifying bins* (Front/Mid/Back), *DriverForm*, *TeamStrength*, *DNF Rates*, and *Overtake Index*. The problem was that the qualifying position signal was low due to it being done through bins meaning might have lost some variation due to grouping them. The features weren't efficiently engineered. Along with this we only ran the model for one season. This model exhibited low feature complexity. Our model also was trained and tested through a manually chosen training and testing percentage (70% and 30% respectively).

5.4.2 Multi-Season PL (2022-2025, 5-fold cross validation)

Expanded to multi-season data with 5-fold grouped cross-validation. Significantly improved generalization by averaging performance across multiple held-out race subsets, reducing the sensitivity to individual season conditions.

5.4.3 PL Enriched Feature (2022-2025, 5-fold cross validation)

Added the full enriched feature set including QualiPercentile, QualiGapToPole, interaction terms, and race-context variables. Both models see moderate improvements with this enhanced feature engineering.

6. Results

6.1 Feature Importance

Feature	Importance
QualiPercentile	0.526981
Quali_x_Overtake	0.113095
DriverForm	0.082553
QualiGapToPole	0.054978
TeamStrength	0.054588
GridSpread	0.019504
OvertakeOpportunity	0.016833
TeammateQualiGap	0.015124
DNFProbFeature	0.013527
Reliability_x_Rain	0.013305
AirTemp	0.012682
TeamDNFRate	0.012416
TrackTemp	0.011675
FP_LongRunPace	0.011600
FP_LongRunVar	0.011173
Driver_vs_Team	0.010585
Rainfall	0.008020
DriverDNFRate	0.006192
OvertakeIndex	0.005167

Table 6a. XGBRanker v2 mean feature importance across all 5 walk-forward folds (all 18 features + DNFProbFeature).

Top Predictors:

- Qualifying position still is dominant
- Performance indicators give us a meaningful secondary signal
- Reliability is incorporated
- Interaction terms improve model flexibility

6.2 Output Files

Dataset: DEBUGF7_f1_enriched_prediction_dataset_2022_2025.csv

XGB Metrics (all-18): v2_walkforward_summary.csv

XGB Metrics (VIF-15): v2_vif_summary.csv

PL Metrics (all-18): pl_v2_walkforward_summary.csv

PL Metrics (VIF-15): pl_v2_vif_summary.csv

Predictions: v2_predictions.csv

Feature Importance: v2_feature_importance.csv

LOO Results: v2_loo_results.csv

VIF Steps (XGB/PL): v2_vif_steps.csv / pl_v2_vif_steps.csv

7. Model Evaluation Metrics

7.1 What These Metrics Measure — and What They Don't

A common misconception is that high evaluation metrics mean the model has correctly predicted the exact finishing order of a race. **None of the metrics used here, except Winner Accuracy, test for exact positional correctness.** They each measure a different aspect of ranking quality, and understanding precisely what each one captures is essential for interpreting the results honestly.

Getting the exact 20-driver finishing order right in any F1 race is essentially impossible from pre-race features alone. Safety cars, mechanical failures, strategy calls, tire wear, and racing incidents all introduce randomness that no model trained on qualifying data, form, and reliability statistics can anticipate. **What a good model can do is extract the systematic, predictable component of race outcomes** — the signal that exists in the data before the lights go out — while acknowledging that the rest is stochastic.

7.2 Metric Definitions

Spearman Rank Correlation

Spearman correlation measures the **monotonic agreement between predicted and actual rank orderings**. For each race, drivers are ranked by their predicted score and by their actual finishing position; Spearman asks how well these two rank lists agree. A value of 1.0 means the predicted order perfectly matches the actual finish; 0 means no relationship; -1.0 means perfectly inverted. **Spearman does not require exact position matches** — a race where every driver finishes exactly one place off from their predicted position still scores near 1.0 because the relative ordering is preserved. The metric is averaged across all races in the test fold.

Kendall's Tau

Kendall's Tau measures the same rank-order agreement but counts **concordant and discordant pairs** of drivers directly. A concordant pair is one where the model correctly predicts that driver A finishes ahead of driver B. $\text{Tau} = (\text{concordant} - \text{discordant}) / \text{total pairs}$. A value of 0.55 means 55% more concordant pairs than discordant ones. Like Spearman, it does not test exact positions — only pairwise ordering. Kendall is generally more conservative than Spearman and less sensitive to large rank swaps at the extremes.

Top-3 and Top-5 Accuracy

Top-k accuracy measures the **overlap between the set of k predicted top finishers and the set of k actual top finishers**. $\text{Top-3 Accuracy} = |\text{predicted top 3} \cap \text{actual top 3}| / 3$. A score of 0.63 means on average 1.9 of the 3 predicted podium drivers actually finished on the podium. Crucially, **this does not check whether each driver was predicted in the correct position within the top 3** — it only checks whether the right drivers were identified as podium finishers regardless of order. A model that correctly identifies all 3 podium drivers but in the wrong order scores 1.0 on this metric.

Winner Accuracy

Winner Accuracy is the **only strict positional metric**: did the driver assigned the highest predicted score actually win the race? This is a binary correct/incorrect judgment averaged across races. A random model would score $1/20 = 5\%$ on 20-driver fields. The qualifying baseline (predict the pole-sitter wins) scores 60.3% because during the 2022-2024 Red Bull era Verstappen frequently won from pole. Winner Accuracy is the hardest metric to beat above the baseline because it penalises any error at the top of the distribution, including upsets caused by safety cars and strategy that no pre-race model can anticipate.

NDCG@k (Normalized Discounted Cumulative Gain)

NDCG is a ranking quality metric borrowed from information retrieval. It assigns **higher reward for correct predictions near the top of the ranking and gives partial credit for near misses**. A driver predicted P2 when

they actually finished P1 scores higher than a driver predicted P10 when they actually finished P1, because the error at P10 wastes more of the top-ranked positions. NDCG@k evaluates only the top k predicted positions. A score of 1.0 is perfect; the 0.89-0.90 range observed here means the top-ranked predictions are highly concentrated around the actual top finishers, even if the exact order within the top k varies.

7.3 Results

All metrics are averaged across the 5 walk-forward folds. Results are reported for both the Two-Stage XGBRanker v2 and Plackett-Luce v2 models.

Metric	XGB v2 (walk-fwd)	PL v2 (walk-fwd)
Mean Spearman	0.662	0.683
Mean KendallTau	0.524	0.546
Top-3 Accuracy	0.662	0.618
Top-5 Accuracy	0.769	0.766
Winner Accuracy	0.513	0.479
NDCG@3	0.891	0.888
NDCG@5	0.898	0.901
DNF LogLoss	0.707	—
DNF Brier Score	0.254	—

Table 7. Final cross-validated metrics — mean across all 5 walk-forward folds.

Spearman of 0.662 (XGBRanker) and 0.683 (PL) reflects moderately strong rank-order agreement across all test races. Top-3 accuracy of 0.662 (XGB) and 0.618 (PL) means the models correctly identify on average nearly 2 of the 3 podium finishers per race. Winner accuracy of 0.513 (XGB) and 0.479 (PL) — the strictest metric — compares against a random baseline of 5% (1-in-20 drivers) and a qualifying baseline of 60.3%. NDCG@3 of 0.891 / 0.888 confirms the highest-ranked predictions cluster tightly around the actual top finishers.

7.4 Why These Results Validate the Model

Since the metrics do not confirm exact positional accuracy, the question of whether the model is genuinely learning — rather than being lucky or overfitting — rests on three independent checks.

Walk-forward validation on truly unseen seasons. The model trains on past seasons and is evaluated on future seasons it has never seen. When the 2022-trained model achieves Spearman 0.65 on 2023 races, or the 2022+2023+2024-trained model scores on 2025 races, those test seasons contain drivers, team performance levels, and race outcomes that did not exist in the training data. Consistent performance across five such folds, covering three different test years, rules out overfitting to any single season.

Beating the qualifying baseline on rank correlation. The qualifying baseline — predicting race finish = qualifying order — requires no training, no features, and no model. It achieves Spearman 0.679 purely because grid position correlates with finishing position in Formula 1. Both the XGBRanker (0.662) and PL (0.683) match or exceed this on Spearman, and the PL feature-selected models (BIC: 0.692, AIC: 0.694) outperform it consistently across folds. Beating a strong domain-knowledge baseline on held-out data is the clearest evidence the model has extracted real signal.

Coefficient signs match domain knowledge. QualiPercentile carries a large negative coefficient — better qualifying means higher predicted strength. DriverForm is positive — recent good results improve the prediction. TeamDNFRate is positive — higher failure rate hurts the driver's expected finish. These directions are exactly what F1 domain knowledge predicts. A model fitting noise would produce random or contradictory coefficient signs; consistent, interpretable signs across all five walk-forward folds confirm the model is capturing genuine

relationships in the data.

7.5 Per-Fold Breakdown and Rolling vs Expanding Comparison

Averaging across folds gives an overall picture, but examining each fold individually reveals how performance varies by season and answers a specific design question: **does training on more historical seasons improve predictions compared to training on only the most recent season?**

The five folds include two pairs where the same test year is evaluated under different training configurations. **For 2024:** Fold 2 trains on 2023 only (rolling) versus Fold 4 which trains on 2022+2023 (expanding). **For 2025:** Fold 3 trains on 2024 only (rolling) versus Fold 5 which trains on 2022+2023+2024 (expanding). Fold 1 (train 2022, test 2023) has no expanding counterpart — it would require pre-2022 data which is not available — so it appears once and contributes to the five-fold total without duplication.

Train	Test	Type	XGB Spearman	XGB Top3	XGB Winner	PL Spearman	PL Top3	PL Winner
2022	2023	Rolling	0.6246	0.5758	0.7727	0.6501	0.5758	0.7273
2023	2024	Rolling	0.6884	0.6528	0.4167	0.7412	0.5833	0.4583
2024	2025	Rolling	0.6459	0.7222	0.3333	0.6512	0.6944	0.3750
2022+2023	2024	Expanding	0.6989	0.6250	0.5000	0.7301	0.5694	0.3750
2022+2023+2024	2025	Expanding	0.6526	0.7361	0.5417	0.6445	0.6667	0.4583

Table 7b. Per-fold Spearman, Top-3, and Winner accuracy for XGBRanker and PL. Shaded rows are the expanding folds (more training data).

Does more historical data help?

Comparing the shaded expanding folds against the rolling folds for the same test year gives a direct answer.

For 2024 predictions (Fold 2 vs Fold 4): adding 2022 data to the 2023 training set consistently improves both models. XGBRanker Spearman rises from 0.688 to 0.699 and Winner Accuracy from 0.417 to 0.500. The PL model similarly improves from Spearman 0.730 to 0.741. More historical data helps because 2022 and 2023 shared a similar competitive structure (Red Bull dominance), so the older season reinforces the same patterns the model needs to predict 2024.

For 2025 predictions (Fold 3 vs Fold 5): the picture is more mixed. XGBRanker gains meaningfully from the full training set — Spearman rises slightly (0.646 to 0.653) and Winner Accuracy jumps from 0.333 to 0.542, suggesting the broader historical context helps the tree model identify likely winners. The PL model shows a smaller Spearman gain but its Winner Accuracy actually falls (0.458 to 0.375). This reflects the **2025 competitive reset**: 2022-2024 was dominated by Red Bull, but 2025 saw Ferrari, McLaren, and Mercedes winning races. The PL model, being a linear ranking model, is more sensitive to this regime shift — including three years of Red Bull-era data can mislead it on a season where that hierarchy no longer holds.

Overall, the comparison suggests that **more data generally helps for stable competitive eras, but a single recent season can be more informative when the sport undergoes a significant competitive reset**. This is a fundamental challenge of sports prediction: historical patterns are only useful when the underlying dynamics have not changed.

8. Comparison and Interpretation

8.1 Performance Comparison Overview

These results show a **clear initial growth followed by diminishing returns**.

8.2 Impact of Data and Evaluation

Metric	Original PL (2024 only)	PL v1 (2022-25, 5-fold)	XGB v1 (2022-25, 5-fold)	PL v2 (walk-fwd)	XGB v2 (walk-fwd)
Mean Spearman	0.535	0.641	0.653	0.683	0.662
Mean KendallTau	0.350	0.508	0.523	0.546	0.524
Top-3 Accuracy	0.571	0.659	0.677	0.618	0.662
Top-5 Accuracy	—	0.717	0.750	0.766	0.769
Winner Accuracy	—	0.447	0.590	0.479	0.513
NDCG@3	—	0.870	0.887	0.888	0.891
NDCG@5	—	0.879	0.890	0.901	0.898
Feature Complexity	Low	Medium	Medium	High	High
Model Type	Linear Ranking	Linear Ranking	Nonlinear ranking	Linear Ranking	Nonlinear ranking

Table 8a. Impact of data coverage and evaluation methodology. v1 = original 5-fold random grouped CV; v2 = walk-forward CV.

The largest improvement happens from moving from the **original single-season (2024) model to the multi-season (2022-2025), cross-validated model**. Expanding the **dataset from 23 to 92 races and along with that switching from a manually chosen single train-test split significantly stabilized the evaluation metrics**. Utilizing 5-fold CV, provides a more stable, less biased estimate of model performance by averaging results across multiple subsets.

Our original plackett-luce model **suffered from high variance due to a limited amount of data**. This made it quite **sensitive to race conditions** and variance from the 2024 season. Now, with **multi-season data and cross-validation**, the model is able to **learn more generalizable patterns** across different circuits. This shows us that the **majority of the performance gain actually comes from improved data coverage and evaluation methodology**.

8.2.1 Feature Importance / Coefficient Summary

Plackett-Luce

The table below shows mean PL coefficients across all 5 walk-forward folds, filtered to features with a mean |coefficient| above 0.03 (excluding degenerate near-zero estimates for unconstrained weather/track features). A negative coefficient means the feature *reduces* race utility (higher QualiPercentile = worse starting position = worse expected finish). A positive coefficient means the feature increases race utility.

Feature	Mean Coefficient (across 5 folds)	Direction
QualiPercentile	-1.6929	↓ hurts finish
DriverForm	0.3008	↑ boosts finish
TeamStrength	0.1652	↑ boosts finish
QualiGapToPole	0.0884	↑ boosts finish
Reliability_x_Rain	0.0753	↑ boosts finish

Feature	Mean Coefficient (across 5 folds)	Direction
OvertakeOpportunity	-0.0627	↓ hurts finish
Quali_x_Overtake	0.0626	↑ boosts finish
TeammateQualiGap	-0.0607	↓ hurts finish
Driver_vs_Team	-0.0548	↓ hurts finish
TeamDNFRate	0.0468	↑ boosts finish

Table 8c. PL v2 mean coefficient summary — only features with mean |coef| ≥ 0.03 shown. Degenerate features (AirTemp, TrackTemp, etc.) are omitted as their coefficients collapse to ~0 when the L-BFGS-B optimiser cannot estimate them from the small training windows.

Two-Stage XGBRanker

Feature importance is measured by mean gain across all trees in all 5 walk-forward folds. See Table 6a for the complete importance ranking. QualiPercentile accounts for 52.7% of total importance — qualifying position remains the dominant signal. Quali_x_Overtake (11.3%) and DriverForm (8.3%) are the next most important features.

Interpretation: These two figures highlight a clear difference in how each model utilizes the predictors. In the Plackett-Luce model, **qualifying position features dominate** strongly. This tells us that the starting position of drivers is the primary predictor in race outcome with **TeamStrength** and **DriverForm** giving small adjustments. The two-stage model however, **distributes importance across features**. DriverForm and TeamStrength actually are more influential than pure qualifying predictors.

8.3 Impact of Feature Engineering (Optimal feature tuning)

Metric	PL v2 (all-18, walk-fwd)	XGB v2 (all-18, walk-fwd)
Mean Spearman	0.683	0.662
Mean KendallTau	0.546	0.524
Top-3 Accuracy	0.618	0.662
Top-5 Accuracy	0.766	0.769
Winner Accuracy	0.479	0.513
NDCG@3	0.888	0.891
NDCG@5	0.901	0.898
Feature Complexity	High	High
Model Type	Linear Ranking	Nonlinear ranking

Table 8b. Comparison of enriched-feature v2 models (walk-forward CV, all 18 features).

The next stage presents **enhanced feature engineering**. A couple key additions include **continuous qualifying representation through QualiPercentile**, qualifying gap features such as **QualiGapToPole** and **TeammateQualiGap**, interaction terms such as **QualiXOvertake**, and contextual race features as well. These features improve the resolution of qualifying performance and are able to **capture relationships between grid position and track characteristics**. Due to this, both models see **moderate improvements**.

8.4 Model Comparison

Despite the complexity of the two-stage approach, **performance gains are minimal**. The two-stage model slightly outperforms in Top-3 and winner accuracy. This suggests that this **model is more focused on identifying top positions**. On the other hand, the **PL model gives us a more consistent overall ranking performance due to its direct likelihood-based optimization**, and slightly outperforms in other metrics such

as Spearman, KendallTau, and top-5 accuracy.

8.5 Why Performance Plateaus

8.5.1 Limited Nonlinearity

The two-stage model is designed to **capture nonlinear interactions using XGBoost**. However, **pre-race variables such as DriverForm and TeamStrength evolve gradually**, and external factors such as weather introduce limited variation. Along with this **qualifying position related predictors capture a majority of the prediction power**. Due to this, there is insufficient nonlinear structure for the model to exploit, leading to minimal performance gains over the simpler PL model. The PL model assumes a linear utility structure and with qualifying position being over half of the importance and a statistically significant predictor, **not being able to capture nonlinear interactions doesn't serve as a detriment to the PL model**.

8.5.2 Diminishing Returns

Once core signals such as qualifying and basic performance metrics are included, additional feature engineering yields smaller improvements. This reflects diminishing returns, where most of the predictive power has already been captured.

8.6 Differences in Model Behavior

Although the overall performance is similar, the models differ in how they make predictions.

Plackett-Luce:

The PL model **optimizes likelihood over full ranking**. From this it is able to provide **consistent global ordering**. This gives it slightly higher Spearman and Kendall.

Two-Stage Model:

This uses LambdaMART to **optimize ranking metrics such as NDCG**. It **emphasizes the top positions** giving it a better top-3 and winner accuracy.

8.7 Main Takeaways

Based on these results there is a clear hierarchy of factors that influence model performance.

1. Evaluation methodology (train-split) and data size
2. Feature engineering
3. Model complexity

Overall, **predictive performance in pre-race F1 modeling is primarily driven by the quality of the features and the overall framework**. Once a strong set of features are established, even simpler models like PL achieve performance that is comparable to more complex ML models that only results in minimal gain.

9. Feature Correlation

9.1 Overall Feature Correlation Analysis

We took all predictors and computed pairwise correlations. Filtered and printed those $\text{abs}(\text{corr}) \geq 0.7$. This was done to identify relationships and potential redundancy between features.

Feature A	Feature B	Correlation
QualiPercentile	Quali_x_Overtake	0.890
QualiPercentile	OvertakeOpportunity	-0.887
DriverForm	TeamStrength	0.833
Rainfall	Reliability_x_Rain	0.820
AirTemp	TrackTemp	0.710

Table 9a. Feature pairs with $|\text{correlation}| \geq 0.7$ (v2 dataset, 2022-2025 seasons).

Strong redundancy among engineered features

Several engineered features show high correlations. This indicates that they actually aren't introducing new information but are **just transformations of existing variables**. An example of this is *QualiXOvertake* and *OvertakeOpportunity* are both highly correlated with *QualiPercentile* (≈ 0.90). This is expected as both of these features are directly constructed using *QualiPercentile*. These features reinforce qualifying performance rather than providing independent predictive signals which suggests there is **redundancy within the feature set**.

Driver and Team Performance Related

There also exists a strong positive correlation between *DriverForm* and *TeamStrength* (≈ 0.82). This reflects how F1 is structured in the real world, where **stronger teams tend to have stronger drivers**. Along with that these drivers tend to be stronger due to the car being better compared to others. While not fully redundant, these **features capture overlapping aspects of performance**, and may partially duplicate information.

Weather-related Variables Correlated

Correlations between *AirTemp* and *TrackTemp* (≈ 0.70) and *Rainfall* and *Reliability_x_Rain* (≈ 0.79) show up as well. These relationships are expected due to **track temperature naturally being dependent on air temperature**, and the feature engineered variable *Reliability_x_Rain* being constructed with the *Rainfall* variable. Due to this, these correlations represent feature construction rather than new meaningful relationships.

Overall

This correlation structure emphasizes that multiple engineered features, especially the interaction terms, are highly dependent on the base variables used for construction. This suggests that even **though feature engineering increases model complexity, it may not proportionally increase independent information**, mainly for qualifying related features.

9.2 Correlation with QualiPercentile

Feature	Corr with QualiPercentile
Quali_x_Overtake	0.890
OvertakeOpportunity	-0.887
DriverForm	-0.625
TeamStrength	-0.619
QualiGapToPole	0.384

Feature	Corr with QualiPercentile
TeamDNFRate	0.237
TeammateQualiGap	0.212
FP_LongRunVar	-0.152
Driver_vs_Team	-0.145
DriverDNFRate	0.086
Reliability_x_Rain	0.061
FP_LongRunPace	-0.011
OvertakeIndex	0.003
Rainfall	0.003
GridSpread	0.002
AirTemp	0.001
TrackTemp	-0.001

Table 9b. Pairwise correlation of each feature with QualiPercentile (v2 dataset, 2022-2025 seasons).

Fixed *QualiPercentile* variable and computed $\text{corr}(\text{QualiPercentile}, \text{feature})$ for all the features. This was to understand how strongly each predictor was correlated with qualifying performance.

Qualifying-derived features dominate

The strongest correlations with *QualiPercentile* are *QualixOvertake* ($\approx +0.90$) and *OvertakeOpportunity* (≈ -0.90). These features are effectively **just re-scaled versions of qualifying performance** and therefore **don't actually introduce new predictive information**. Instead, they increase the importance of qualifying within the model.

Moderate relationships capture additional qualifying context

Features such as *QualiGapToPole* (≈ 0.41) and *TeammateQualiGap* (≈ 0.21) both show moderate relationships. These features are **still related to qualifying but not perfectly correlated**. This means they provide additional nuance — information that is relative performance within a team or field gaps. The correlations tell us that these features are more useful than purely derived features.

Driver and Team Performance Inversely Related to Qualifying

The features *DriverForm* (≈ -0.63) and *TeamStrength* (≈ -0.63) show an inverse relationship with qualifying position. What this means is that **better drivers and teams tend to qualify in stronger positions (lower percentile values)**. This then results in a negative correlation. This tells us that these features also contribute meaningful performance information beyond qualifying alone.

Independent features of Qualifying

There are a couple near zero correlations such as weather features including *AirTemp*, *TrackTemp*, and *Rainfall*, track characteristics such as *OvertakeIndex*, and race variability in *GridSpread*. These features capture some of the key race-specific or environmental effects that aren't reflected in qualifying. This makes **these features, although weak, a potential source of independent signals**.

Overall

Qualifying performance, as expected, emerges as the dominant predictor with several **feature engineered features effectively duplicating its signal**. However, **driver and team performance metrics provide additional independent information**, while weather and track specific variables remain largely uncorrelated, capturing separate aspects of race variability.

10. Plackett-Luce Model Selection Criteria

10.1 Motivation

Unlike the **two-stage ML model**, the **Plackett-Luce model gives a full statistical inference, including coefficient estimates, standard errors, z-scores, and p-values**. This allows for the use of formal model selection techniques that balance model fit and complexity. Later on, a leave-one-out (LOO) method will be used for the two-stage model to evaluate predictive performance, but this method doesn't explicitly penalize unnecessary model complexity. This is why different techniques are required for the Plackett-Luce model.

10.2 Model Selection Methods

Several model selection techniques were considered to evaluate the contribution of the different predictors in the Plackett-Luce model.

10.2.1 Akaike Information Criterion (AIC)

AIC evaluates model quality by balancing goodness of fit with model complexity: **$AIC = -2 \log L + 2k$** where **L is the likelihood and k is the number of parameters**. This approach encourages good model fit, allows for more flexibility in retaining predictors, and is also well-suited for predictive performance. On the other hand, it **penalizes complexity less strongly and may retain redundant features when predictors are highly correlated**.

10.2.2 Bayesian Information Criterion (BIC)

BIC is similar to AIC but applies a stronger penalty for model complexity: **$BIC = -2 \log L + k \log n$** where **n is the number of observations**. As mentioned, BIC incorporates a **stronger penalty** for unnecessary features, is **more effective at removing redundancy**, and tends to produce simpler, more interpretable models. It may remove features that provide small but meaningful improvements and can be **overly conservative** in some cases.

10.2.3 Forward Selection

Forward selection begins with an empty model and iteratively adds features that improve the selection criterion whether it be AIC or BIC. This method is simple to implement and builds the model progressively. The problem is that it **can miss important features combinations and is sensitive to correlated predictors**.

10.2.4 Backward Selection

Backward selection begins with the full model and iteratively removes the least useful features based on a chosen criterion (AIC or BIC). This starts with all the available information, is **more efficient when features are correlated**, and is well-suited when an initial full model is complete. Computationally though, this may be more expensive and may still be affected by multicollinearity.

10.2.5 Stepwise Selection (Forward-Backward)

Stepwise selection combines both forward and backward approaches by iteratively adding and removing features. This can be more flexible than purely forward or backward selection and can better navigate correlated feature spaces. Some cons include that it is more complex to implement, and is still heuristic in nature.

10.3 Method Selection

Due to the nature of the enriched feature set, which as we have come to know, contains several highly correlated predictors, model selection methods that effectively handle redundancy are preferred.

10.3.1 Primary Method: BIC + Backward Selection

This approach was chosen due to the **feature set containing substantial correlation and redundancy**, **BIC strongly penalizing unnecessary complexity**, and **backward selection being a well-suited method to refine an already fully completed model**.

10.3.2 Secondary Method: AIC + Stepwise Selection

This approach is used as a complementary method because AIC allows slightly more flexibility in retaining predictors and stepwise selection can explore a broader range of feature combinations. This serves as a robustness check against the more conservative BIC approach.

10.4 BIC + Backward Selection

10.4.1 Methodology

This process starts with the full Plackett-Luce model containing all of the predictors. Using the full dataset, the model is fit by maximizing the penalized likelihood with L2 regularization. The **log-likelihood of the fitted model is used to compute the BIC value which serves as the baseline score**.

At **each iteration, one predictor is removed at a time, and the model is refitted with the reduced predictor set**. For each refitted model, the log-likelihood is calculated and converted to a BIC score. The **predictor whose removal results in the largest decrease of BIC is permanently removed from the model**. This process is repeated iteratively until no further reduction in BIC can be achieved.

After obtaining the final set of predictors, the Plackett-Luce model is retrained using those features and **evaluated using walk-forward cross-validation** with the same 5 folds used throughout this study. Selection is performed on 2022-2024 reference data only, so the 2025 test fold is never seen during feature selection. The v2 implementation corrects a bug from v1 where BIC used n = number of driver-race rows instead of n = number of races; the corrected formula uses **$BIC = -2 \log L + k \log(n_{\text{races}})$** .

10.4.2 Selection Steps

Starting from the full 18-feature model (BIC = 5366.974), 14 features were removed across 14 steps. The process stopped when no further removal improved BIC.

Step	Feature Removed	BIC Before	BIC After	BIC Delta	Features Left
1	FP_LongRunPace	5366.974	5362.745	-4.229	17
2	TrackTemp	5362.745	5358.525	-4.220	16
3	Rainfall	5358.525	5354.306	-4.220	15
4	AirTemp	5354.306	5350.086	-4.220	14
5	GridSpread	5350.086	5345.867	-4.220	13
6	OvertakeIndex	5345.867	5341.647	-4.220	12
7	TeamDNFRate	5341.647	5337.477	-4.170	11
8	FP_LongRunVar	5337.477	5334.071	-3.406	10
9	Reliability_x_Rain	5334.071	5330.958	-3.113	9
10	TeamStrength	5330.958	5327.985	-2.973	8
11	Quali_x_Overtake	5327.985	5325.760	-2.225	7
12	DriverDNFRate	5325.760	5323.413	-2.347	6
13	TeammateQualiGap	5323.413	5322.787	-0.626	5
14	QualiGapToPole	5322.787	5320.711	-2.076	4

10.4.3 Selected Features and Coefficients

BIC backward selection retained **4 features** (final BIC = 5320.711). BIC's strong $\log(n)$ penalty is highly aggressive here, pruning 14 of 18 predictors and retaining only the features with the clearest independent signal. Mean PL coefficients across the 5 walk-forward folds are shown below.

Feature	Mean Coef	Direction
QualiPercentile	-1.6529	lower finish score
DriverForm	+0.4280	higher finish score
Driver_vs_Team	-0.1363	lower finish score
OvertakeOpportunity	-0.1257	lower finish score

10.4.4 Walk-Forward Performance

Walk-forward mean metrics for the BIC-selected 4-feature model vs. the qualifying-position baseline:

Method	Spearman	KendallTau	Top3 Acc	Top5 Acc	Winner Acc	NDCG@3	NDCG@5
Model	0.6922	0.5554	0.6260	0.7644	0.4447	0.8873	0.9029
Baseline	0.6790	0.5404	0.6742	0.7727	0.6030	0.9001	0.9123

The BIC-selected model improves over the baseline on rank-correlation metrics (Spearman, KendallTau) but falls short on accuracy-based metrics (Top3, Winner Accuracy). This is consistent with BIC's aggressive pruning: with only 4 features, the model captures the broad competitive order well but lacks the granularity to reliably identify podium finishers and race winners.

10.5 AIC + Forward Stepwise

10.5.1 Methodology

AIC forward stepwise selection starts with an empty model and **greedily adds the one feature whose inclusion gives the largest AIC decrease** at each step. The process stops when no remaining feature strictly reduces AIC. The v2 implementation uses **forward-only selection** to avoid the infinite oscillation that can occur with bidirectional stepwise when correlated features trade off against each other. AIC is computed as **$AIC = -2 \log L + 2k$** where n is the number of races. Selection is performed on 2022-2024 reference data with the same walk-forward evaluation as BIC.

10.5.2 Selection Steps

Starting from an empty model, 5 features were added before AIC stopped improving.

Step	Feature Added	AIC Before	AIC After	AIC Delta	Features
1	QualiPercentile	—	5388.980	—	1
2	TeamStrength	5388.980	5323.465	-65.514	2
3	Quali_x_Overtake	5323.465	5317.430	-6.035	3
4	DriverForm	5317.430	5312.425	-5.006	4
5	TeamDNFRate	5312.425	5310.559	-1.866	5

10.5.3 Selected Features and Coefficients

AIC forward stepwise retained **5 features** (final AIC = 5310.559). AIC's lighter $2k$ penalty allows more features than BIC while still discarding predictors with little independent contribution. Notably, AIC selected TeamStrength and Quali_x_Overtake where BIC selected OvertakeOpportunity and Driver_vs_Team — reflecting AIC's greater tolerance for correlated team-based features.

Feature	Mean Coef	Direction
QualiPercentile	-1.6433	lower finish score
TeamStrength	+0.2501	higher finish score
DriverForm	+0.1790	higher finish score
Quali_x_Overtake	+0.1231	higher finish score
TeamDNFRate	+0.0752	higher finish score

10.5.4 Walk-Forward Performance

Walk-forward mean metrics for the AIC-selected 5-feature model vs. the qualifying-position baseline:

Method	Spearman	KendallTau	Top3 Acc	Top5 Acc	Winner Acc	NDCG@3	NDCG@5
Model	0.6944	0.5559	0.6407	0.7694	0.4356	0.8899	0.9017
Baseline	0.6790	0.5404	0.6742	0.7727	0.6030	0.9001	0.9123

The AIC-selected model shows a similar pattern to BIC: rank-correlation metrics (Spearman 0.694, KendallTau 0.556) exceed the baseline, while accuracy-based metrics (Top3, Winner) remain below it. The 5-feature AIC model is marginally stronger than the 4-feature BIC model across most metrics, suggesting the one additional feature (TeamStrength) provides incremental predictive value.

10.6 Variance Inflation Factor (VIF) Analysis

10.6.1 What is VIF and Why Do We Use It?

The **Variance Inflation Factor (VIF)** measures **multicollinearity** — the degree to which a predictor can be linearly explained by the other predictors in the model. For a given feature x_j , VIF is computed by regressing x_j on all remaining features and calculating:

$$\text{VIF}(x_j) = 1 / (1 - R_j^2)$$

where R_j^2 is the coefficient of determination from that regression. A VIF of 1 means no multicollinearity — the feature carries information not present in any other predictor. A VIF of 5 means 80% of that feature's variance is explained by the other features, indicating strong redundancy. The standard threshold of **VIF < 5** (equivalently $R^2 < 0.8$) is used throughout this study.

Why multicollinearity is a problem: When two or more features are highly correlated, they carry overlapping information. This creates two distinct problems depending on the model type:

For **linear probabilistic models like Plackett-Luce**, coefficients are estimated via maximum likelihood. When features are collinear, the Fisher information matrix — which determines coefficient standard errors and p-values — becomes near-singular. This inflates standard errors, making features appear statistically insignificant even when they carry real signal, and invalidates the Wald test p-values used for inference.

For **tree-based models like XGBRanker**, multicollinearity does not directly bias predictions the way it does in linear models. However, correlated features split feature importance scores across multiple redundant predictors — if two features convey the same information, the model may use either one interchangeably across trees. This makes feature importance rankings unreliable and masks which features are truly driving predictions.

The VIF backward selection procedure addresses both problems by iteratively removing the most collinear feature until all remaining features are below the threshold:

Compute VIF for all features simultaneously on the 2022-2024 reference dataset.

Remove the single feature with the highest VIF.

Recompute VIF for the remaining features.

Repeat until all VIFs fall below 5.0.

An important property of VIF is that it depends only on the **feature matrix X** — not on the model type, the outcome variable, or any model-specific parameters. Because both the Plackett-Luce model and XGBRanker use the same 18 engineered features on the same 2022-2024 reference dataset, the VIF selection procedure produces an **identical removal sequence and identical retained feature set for both models**. The redundancy structure of the features is entirely model-independent.

10.6.2 Plackett-Luce Model

Why VIF for PL: The Plackett-Luce model estimates a coefficient vector β via maximum likelihood; standard errors and p-values are derived from the Fisher information matrix $F = -\partial^2 \log L / \partial \beta^2$. When predictors are collinear, F becomes near-singular and its inverse — the asymptotic covariance matrix — has inflated diagonal entries, leading to unreliable standard errors. VIF selection ensures F is well-conditioned, giving valid frequentist inference for the retained coefficients.

How it is implemented: VIF backward selection is run on the 2022-2024 reference dataset (68 races, 1,836 driver-race rows), starting from the full set of 18 features. Three features are removed in sequence:

Step	Feature Removed	VIF at Removal	Features Left
1	OvertakeOpportunity	Inf	17
2	DriverForm	613.93	16
3	Quali_x_Overtake	18.04	15

Table 10a. VIF backward selection steps for PL model (threshold = 5.0, reference: 2022-2024 data).

OvertakeOpportunity is removed first because it is defined as $(1 - QualiPercentile) \times OvertakeIndex$ — a deterministic product of two other features already in the set, making its VIF infinite. *DriverForm* follows because $Driver_vs_Team = DriverForm - TeamStrength$, so *DriverForm* is nearly entirely predictable from the other two. *Quali_x_Overtake* is removed last due to its strong linear relationship with the qualifying features. After these three removals, all 15 remaining features have VIF below 5.0.

Feature	VIF
Reliability_x_Rain	3.924
Rainfall	3.762
AirTemp	2.200
TrackTemp	2.052
QualiPercentile	2.045
QualiGapToPole	1.935
TeamStrength	1.885
TeamDNFRate	1.524
GridSpread	1.432
TeammateQualiGap	1.428
FP_LongRunPace	1.418
DriverDNFRate	1.342
FP_LongRunVar	1.228
OvertakeIndex	1.212
Driver_vs_Team	1.195

Table 10b. Final VIF values for the 15 retained PL features (all below the 5.0 threshold).

Evaluation metrics — PL VIF-15 vs All-18:

Metric	PL All-18	PL VIF-15	Delta (VIF – All-18)
Mean Spearman	0.683	0.685	+0.0013
Mean KendallTau	0.546	0.546	+0.0003
Top-3 Acc	0.618	0.620	+0.0025
Top-5 Acc	0.766	0.768	+0.0017
Winner Acc	0.479	0.462	-0.0167
NDCG@3	0.888	0.890	+0.0019
NDCG@5	0.901	0.902	+0.0016

Table 10c. All-18 vs VIF-selected PL model (walk-forward means).

Removing three collinear features leaves predictive performance essentially unchanged, confirming they were statistically redundant. The slight metric shifts are within noise. The retained 15-feature set gives more reliable coefficient estimates and valid p-values for PL inference, which is the primary purpose of VIF selection for a linear probabilistic model.

10.6.3 XGBRanker

Why VIF for XGBRanker: Gradient-boosted trees can handle correlated features without biased predictions — the model will not produce misleading coefficients the way a linear model does. However, when two features are highly correlated, XGBoost may use either one across different trees, artificially splitting the importance score between them. This makes feature importance rankings hard to interpret: a feature may appear unimportant simply because its signal was absorbed by a correlated partner. VIF selection consolidates the feature set to the maximally non-redundant subset, producing importance scores that reflect genuine independent contributions.

How it is implemented: The same backward VIF procedure is applied to the same 2022-2024 reference dataset. Because VIF is a property of the feature matrix alone, the removal sequence is identical to the PL case: OvertakeOpportunity (Inf), DriverForm (613.93), and Quali_x_Overtake (18.04) are removed in that order, leaving 15 features.

Step	Feature Removed	VIF at Removal	Features Left
1	OvertakeOpportunity	Inf	17
2	DriverForm	613.93	16
3	Quali_x_Overtake	18.04	15

Table 10d. VIF backward selection steps for XGBRanker v2 (threshold = 5.0, reference: 2022-2024 data).

The 15 retained features span all feature groups: three qualifying metrics (QualiPercentile, QualiGapToPole, TeammateQualiGap), team/driver performance (TeamStrength, Driver_vs_Team), practice pace and consistency (FP_LongRunPace, FP_LongRunVar), reliability (DriverDNFRate, TeamDNFRate, Reliability_x_Rain), environmental (AirTemp, TrackTemp, Rainfall), and track characteristics (OvertakeIndex, GridSpread). Each retained feature carries information not linearly explained by the others.

Evaluation metrics — XGBRanker VIF-15 vs All-18:

Metric	XGB All-18	XGB VIF-15	Delta (VIF – All-18)
Mean Spearman	0.662	0.662	-0.0003
Mean KendallTau	0.524	0.519	-0.0051

Metric	XGB All-18	XGB VIF-15	Delta (VIF – All-18)
Top-3 Acc	0.662	0.652	-0.0109
Top-5 Acc	0.769	0.780	+0.0118
Winner Acc	0.513	0.486	-0.0273
NDCG@3	0.891	0.890	-0.0007
NDCG@5	0.898	0.904	+0.0058

Table 10e. All-18 vs VIF-selected XGBRanker (walk-forward means).

Overall ranking performance is preserved after VIF reduction. Spearman and Kendall's Tau remain nearly identical, confirming that removing three redundant features does not reduce the model's ability to order drivers correctly. Top-3 and Top-5 accuracy show marginal changes within fold-to-fold noise. The primary gain from VIF selection for XGBRanker is interpretability: feature importance scores for the 15-feature model reflect genuine independent contributions rather than being diluted across correlated partners.

10.7 Overall

We see that the **BIC plus backward selection model achieves comparable or slightly improved performance while only using 4 instead of the original 18 predictors** (v1 result; v2 pending). This tells us that a **majority of the feature-engineered features were redundant, qualifying and core performance metrics contain most of the predictive signal**, and removing **weaker features reduces noise**. These results show that a parsimonious model (model that explains data with fewer or least amount of features), based on a core set of predictors can match or outperform a highly engineered feature set.

10.7 Cross-Method Comparison: VIF vs BIC vs AIC

The three PL model selection methods — VIF backward selection, BIC backward selection, and AIC forward stepwise — each approach feature reduction from a different angle. VIF removes features purely on multicollinearity grounds without regard to predictive performance. BIC and AIC balance goodness of fit against complexity, with BIC applying a heavier penalty. The table below compares the mean walk-forward performance of each selected model against the full 18-feature PL model and the qualifying-position baseline.

Model	Feats	Spearman	Kendall	Top3	Top5	Winner	NDCG@3	NDCG@5
Full PL (18 features)	18	0.6834	0.5456	0.6179	0.7661	0.4788	0.8879	0.9007
PL + VIF (15 features)	15	0.6847	0.5459	0.6205	0.7677	0.4621	0.8898	0.9023
PL + BIC (4 features)	4	0.6922	0.5554	0.6260	0.7644	0.4447	0.8873	0.9029
PL + AIC (5 features)	5	0.6944	0.5559	0.6407	0.7694	0.4356	0.8899	0.9017
Qualifying Baseline	1	0.6790	0.5404	0.6742	0.7727	0.6030	0.9001	0.9123

Rank correlation (Spearman, Kendall): All PL models beat the baseline, and feature reduction consistently improves rank correlation — AIC (5 features) achieves the best Spearman (0.694) and Kendall (0.556) of any model. This pattern — fewer features yielding better rank correlation — indicates the dropped features were adding noise, not signal, to the overall ordering.

Accuracy metrics (Top3, Top5, Winner): The qualifying baseline outperforms all models on these metrics. Among the PL variants, AIC achieves the highest Top3 accuracy (0.641), while the full model and VIF model are competitive. BIC and AIC both drop Winner Accuracy below the full model — aggressive pruning removes features that help identify the exact race winner even if they hurt overall rank ordering.

Key takeaway: The three selection methods serve different purposes. VIF corrects multicollinearity with minimal feature loss (18 → 15), preserving predictive breadth. BIC produces the most parsimonious model (4 features)

with the strongest interpretability. AIC strikes the best overall balance — 5 features, best rank correlation, and competitive accuracy — making it the preferred reduced model for the PL framework.

10.8 McFadden's Adjusted Pseudo-R² and Coefficient Analysis

10.8.1 Methodology

What is R² and why can't we use it directly?

In linear regression, R² (the coefficient of determination) measures the proportion of variance in the outcome that the model explains: $R^2 = 1 - SS_{\text{residual}} / SS_{\text{total}}$, where SS_{total} is the total variance of the response and SS_{residual} is the unexplained variance after fitting. A value of 0 means the model explains nothing, 1 means it explains everything. R² is important because it gives an intuitive, scale-free summary of how well the model fits the data.

The Plackett-Luce model is not a regression model — it does not predict a continuous outcome or minimize a sum of squared residuals. Instead, it maximizes the likelihood of observed race orderings. There is no sum of squares to partition, so standard R² has no direct meaning here. **A goodness-of-fit measure that works with likelihoods is needed.**

What is McFadden's Pseudo-R² and why do we use it?

McFadden's pseudo-R² is the standard goodness-of-fit measure for likelihood-based models (logistic regression, Plackett-Luce, etc.). It is constructed by analogy with R²: instead of comparing residual variance to total variance, it compares the log-likelihood of the fitted model to the log-likelihood of a **null model** that contains no predictors at all.

$$\text{McFadden } R^2 = 1 - \log L_{\text{model}} / \log L_{\text{null}}$$

The **null model** for the Plackett-Luce framework assigns equal probability to every driver at every sequential selection step — equivalent to saying every possible race ordering is equally likely. Its log-likelihood is $\log L_{\text{null}} = -\sum \log(k_i!)$ summed over all races, where k_i is the number of starters in race i . When the fitted model's log-likelihood equals the null (no improvement), $R^2 = 0$. As the model explains more of the ordering structure, $\log L_{\text{model}}$ approaches 0 (from below) and R^2 approaches 1. For sports ranking models, values of 0.05–0.20 indicate meaningful predictive power — the scale is lower than in regression because even the best pre-race model cannot fully predict the chaotic dynamics of a live race.

What is the Adjusted R² and why does it matter?

A well-known problem with R² in any model is that it **never decreases when you add more predictors**, even if those predictors are pure noise. Adding a random feature to a model will always increase (or at worst hold constant) the R², because more parameters allow the model to fit idiosyncratic patterns in the training data. This makes raw R² a poor criterion for comparing models with different numbers of features.

Adjusted R² corrects for this by penalizing each additional parameter. For McFadden's pseudo-R², the adjusted version subtracts the number of model parameters k from the log-likelihood before computing the ratio:

$$\text{McFadden Adjusted } R^2 = 1 - (\log L_{\text{model}} - k) / \log L_{\text{null}}$$

Subtracting k from $\log L_{\text{model}}$ makes the numerator smaller (more negative), which shrinks the ratio and reduces R². A new feature only improves adjusted R² if its improvement in log-likelihood exceeds its parameter cost. This makes adjusted R² the correct metric for comparing the Full (18 features), VIF (15), BIC (4), and AIC (5) models — it directly answers: **does adding these extra features improve the model enough to justify their complexity?**

How were the coefficients and p-values computed?

Each of the four PL model variants is fit a single time on the **2022-2024 reference dataset** — that is, all 68 races from the 2022, 2023, and 2024 seasons combined into one training set. This is distinct from the walk-forward cross-validation used elsewhere in this study. In walk-forward CV, we repeatedly split data into training and test years to measure predictive performance on unseen seasons. Here, the goal is different: we want to measure **goodness of fit on the training data** and obtain stable coefficient estimates for inference. The 2025 season is never used in any part of this analysis — not for training, not for testing, not for feature selection. It exists solely as a final out-of-sample evaluation period (tested in the walk-forward folds).

The coefficients themselves come directly from the L-BFGS-B optimizer that maximizes the penalized Plackett-Luce log-likelihood. Each coefficient β_j represents how much a one-unit increase in feature j (after standardization) increases a driver's log-strength, which in turn increases their probability of finishing ahead of rivals.

Standard errors are computed from the **unregularized Fisher information matrix** — the negative Hessian of the plain log-likelihood (without the L2 penalty term) evaluated at the fitted coefficients. The Fisher information measures how sharply the likelihood peaks around the estimates: a steep peak means low uncertainty (small SE), a flat peak means high uncertainty (large SE). Using the unregularized Hessian is necessary for valid frequentist inference because the L2 penalty artificially steepens the curvature, which would understate the true uncertainty. The **Z-score** is then $Z = \beta / \text{SE}$ and the **p-value** is computed from the standard normal distribution as $p = 2 \times (1 - \Phi(|Z|))$, giving a two-sided Wald test.

10.8.2 Goodness-of-Fit Comparison

Null log-likelihood (uniform random ranking, 68 races): $\log L_{\text{null}} = -2869.835$

Model	k	log L	R ²	Adj R ²
Full (18 features)	18	-2644.277	0.0786	0.0723
VIF (15 features)	15	-2651.118	0.0762	0.0710
BIC (4 features)	4	-2649.756	0.0767	0.0753
AIC (5 features)	5	-2648.279	0.0772	0.0755

The raw R² values (0.076–0.079) are similar across all four models. However, the **adjusted R²** tells a different story: the BIC (0.0753) and AIC (0.0755) models achieve **higher adjusted R² than the full 18-feature model (0.0723)**. This confirms that the 13–14 features removed by BIC and AIC were adding parameters without improving fit. The VIF model (0.0710) has the lowest adjusted R², reflecting that VIF selects on collinearity alone and retains some features whose marginal likelihood contribution is minimal.

Interpreting the 7.7% figure: why the models still perform well

A McFadden R² of ~0.077 may appear low, but it is important to understand what the null model actually is. The null assigns **equal probability to every possible finishing order**. For a 20-driver race, there are $20! \approx 2.4 \times 10^{18}$ possible orderings. The null treats each of these as equally likely — an astronomically uninformed baseline. The 7.7% is the relative improvement in log-likelihood over this completely random lottery, not the proportion of variance explained in the familiar regression sense.

The high evaluation metrics (Spearman ≈ 0.69 , NDCG@3 ≈ 0.89) exist for a different reason: **Formula 1 race results are inherently structured**. The same drivers and constructors consistently occupy the same part of the field across races. A model that has been trained on historical races learns this structure — which drivers tend to qualify and finish at the front, which teams have the fastest car, which drivers are in form. The evaluation metrics measure how well the model captures that learned structure when predicting future, unseen races. They are not measuring improvement over a random baseline; they are measuring agreement with actual observed rankings.

The two measures are therefore answering completely different questions. McFadden R^2 asks: *compared to knowing nothing at all, how much does the model improve?* The evaluation metrics ask: *when the model ranks drivers and we compare to what actually happened, how accurate is the ordering?* A Spearman correlation of 0.69 is achievable even with a small McFadden R^2 because the difficulty of the problem is not as hard as 20! suggests — the competitive hierarchy makes many orderings far more likely than others before any model is applied.

Why the qualifying baseline achieves high Spearman despite race variation

The qualifying baseline (predicting that the race finishes in the same order as qualifying) achieves Spearman $\approx$ 0.679 despite the fact that qualifying and race order never match perfectly. This is not surprising once the properties of Spearman correlation are understood.

Spearman correlation does not require exact position matches — it only requires that the **relative ordering is preserved**. A race where the pole-sitter finishes P2, the P2 starter finishes P1, and every other driver finishes exactly one position off still produces a Spearman near 1.0, because the rank ordering is almost entirely intact. What matters is not whether P3 finishes P3, but whether the driver who started P3 finishes ahead of the driver who started P10.

Formula 1 competitive order operates in **performance bands**. The front runners (top 6-8 drivers across the top 3-4 teams) shuffle among themselves but almost never drop to P15. The midfield (P9-P14) shuffles within its own band. Even the backmarkers (P15-20) tend to shuffle among themselves rather than jumping into the points. This within-band shuffling does relatively little damage to Spearman because the major rank separations across bands are preserved.

DNFs are a genuine source of rank disruption — a driver who qualifies P1 and retires is recorded at the back of the finishing order, creating a large rank jump that pulls correlation down for that race. However, DNFs affect 1–3 drivers per race on average, so while they create noise in individual races they do not dominate the mean Spearman across 68 races.

The 2024 and 2025 seasons were more competitive than 2022-2023 (the Red Bull era), with Ferrari, McLaren, and Mercedes all winning races. This increased inter-team variation lowers the qualifying-to-finish correlation for those years. The mean Spearman of 0.679 reflects an average across all five walk-forward folds — folds testing on the more predictable 2022-2023 seasons contribute higher correlations while the 2025 fold contributes lower, and the 0.679 is the result of averaging across this range.

10.8.3 Coefficients and p-values by Model

Each model is fit once on the 2022-2024 reference data. Coefficients reflect the direction and magnitude of each feature's contribution to a driver's Plackett-Luce 'strength' — a higher score increases the probability of finishing ahead of rivals. Continuous features other than QualiPercentile are standardized, so coefficients are comparable in magnitude. Standard errors and p-values use the unregularized Fisher information (Wald test).

Note on the full and VIF models: L2 regularization collapses several correlated features (AirTemp, TrackTemp, Rainfall, OvertakeIndex, GridSpread) to near-zero coefficients. In those models, multicollinearity inflates standard errors for the remaining features, reducing statistical significance even for genuinely important predictors. The BIC and AIC models avoid this by removing the redundant features, yielding cleaner and more interpretable inference.

BIC-selected model (4 features)

Feature	Coef	SE	Z	p-value	Sig
QualiPercentile	-2.2883	0.4759	-4.808	0.0000	***
DriverForm	+0.4159	0.0450	9.231	0.0000	***

OvertakeOpportunity	-0.3144	0.1532	-2.053	0.0401	*
Driver_vs_Team	-0.1368	0.0346	-3.956	0.0001	***

Significance codes: *** p<0.001 ** p<0.01 * p<0.05 . p<0.1 (SE and p-values from unregularized Fisher information)

AIC-selected model (5 features)

Feature	Coef	SE	Z	p-value	Sig
QualiPercentile	-2.2719	0.4765	-4.768	0.0000	***
TeamStrength	+0.2477	0.0591	4.191	0.0000	***
Quali_x_Overtake	+0.3099	0.1535	2.018	0.0436	*
DriverForm	+0.1633	0.0588	2.775	0.0055	**
TeamDNFRate	+0.0649	0.0337	1.925	0.0543	.

Significance codes: *** p<0.001 ** p<0.01 * p<0.05 . p<0.1 (SE and p-values from unregularized Fisher information)

VIF-selected model (15 features)

Feature	Coef	SE	Z	p-value	Sig
Reliability_x_Rain	+0.0658	0.0606	1.085	0.2779	
Rainfall	-0.0000	—	—	—	
AirTemp	+0.0000	0.0000	2.913	0.0036	**
TrackTemp	-0.0000	0.0000	-1.537	0.1243	
QualiPercentile	-1.4712	0.1475	-9.972	0.0000	***
QualiGapToPole	+0.1143	0.0546	2.094	0.0363	*
TeamStrength	+0.3980	0.0432	9.213	0.0000	***
TeamDNFRate	+0.0346	0.0392	0.882	0.3777	
GridSpread	+0.0000	0.0000	23.510	0.0000	***
TeammateQualiGap	-0.0630	0.0335	-1.877	0.0605	.
FP_LongRunPace	+0.0406	0.0838	0.484	0.6281	
DriverDNFRate	+0.0204	0.0358	0.568	0.5697	
FP_LongRunVar	-0.0274	0.0393	-0.698	0.4853	
OvertakeIndex	-0.0000	0.0000	-19.406	0.0000	***
Driver_vs_Team	+0.0880	0.0330	2.669	0.0076	**

Significance codes: *** p<0.001 ** p<0.01 * p<0.05 . p<0.1 (SE and p-values from unregularized Fisher information)

Full model (18 features)

Feature	Coef	SE	Z	p-value	Sig
QualiPercentile	-2.3428	0.4798	-4.883	0.0000	***
QualiGapToPole	+0.1073	0.0553	1.939	0.0525	.
TeammateQualiGap	-0.0638	0.0339	-1.883	0.0597	.
FP_LongRunPace	+0.0469	0.0840	0.558	0.5770	
FP_LongRunVar	-0.0304	0.0394	-0.772	0.4404	
DriverForm	+0.4639	1.0226	0.454	0.6501	
TeamStrength	-0.0015	0.9075	-0.002	0.9987	

DriverDNFRate	+0.0263	0.0359	0.732	0.4639	
TeamDNFRate	+0.0322	0.0392	0.822	0.4108	
AirTemp	-0.0000	0.0000	-0.057	0.9548	
TrackTemp	-0.0000	0.0000	-0.069	0.9450	
Rainfall	+0.0000	0.0000	0.383	0.7014	
OvertakeIndex	-0.0000	0.0000	-0.143	0.8861	
Quali_x_Overtake	+0.1536	0.0768	2.001	0.0454	*
Reliability_x_Rain	+0.0605	0.0607	0.997	0.3190	
OvertakeOpportunity	-0.1538	0.0769	-2.001	0.0454	*
Driver_vs_Team	-0.1501	0.5397	-0.278	0.7810	
GridSpread	-0.0000	0.0000	-0.314	0.7536	

Significance codes: *** p<0.001 ** p<0.01 * p<0.05 . p<0.1 (SE and p-values from unregularized Fisher information)

10.8.4 Interpretation

QualiPercentile is the strongest and most consistently significant predictor across all four models ($p < 0.001$ in all). Its large negative coefficient (approximately -2.3) means a driver starting from pole (QualiPercentile = 0) has substantially higher predicted strength than one starting from the back (QualiPercentile = 1). This confirms qualifying position as the dominant pre-race signal in Formula 1.

DriverForm is significant in the BIC model ($p < 0.001$) and AIC model ($p = 0.006$), with a positive coefficient indicating that drivers in better recent form are predicted to finish higher. In the full model, its significance is diluted by collinear features sharing the same signal.

TeamStrength is significant in the AIC and VIF models ($p < 0.001$), reflecting the constructor's pace advantage independent of qualifying. BIC did not retain TeamStrength, preferring OvertakeOpportunity and Driver_vs_Team as alternative signals for team-level effects.

The **multicollinearity issue** is most visible in the full model: DriverForm has coef = +0.46 but $p = 0.65$, meaning its variance is shared with correlated features and its individual contribution cannot be isolated. The BIC and AIC selection process resolves this by retaining only features that provide genuinely independent information, which is why the adjusted R^2 improves even as the feature count drops from 18 to 4 or 5.

11. Leave One Out Analysis for Two-Stage Model

11.1 Methodology

To further evaluate the contribution of individual predictors, we utilized a leave one out (LOO) feature analysis. The full model is first trained using all features to establish the baseline model performance. After, **each feature is removed one at a time and the model is retrained, and the resulting evaluation metrics are compared to the baseline**. What this analysis does is **measure the marginal contribution of each feature to model performance**. Unlike feature importance from tree-based models, which reflects how frequently a feature is used in splits, **LOO directly looks at how model performance changes when a feature is removed**.

However, due to many of the features being highly correlated, especially the qualifying derived features, removing a single one, may not actually significantly affect results if the other correlated features can substitute for it. To address this multicollinearity, variance inflation factor (VIF) is computed for each predictor. VIF measures **how much of a feature can be explained by other features**. For a feature X_j , regress it on all features and compute R^2 from that regression:

$$VIF_j = 1 / (1 - R^2_j)$$

From our problem and the correlation analysis we already saw that *QualiPercentile* is highly correlated with its engineered features and *DriverForm* and *TeamStrength* also exhibit high correlation. We can utilize iterative VIF removal, just like backward selection to find the key features. The features with high VIF are highly collinear and can be removed and after each removal the VIF values are recomputed using the new feature set.

11.2 Leave One Out Results

Feature Removed	Δ Spearman	Δ Kendall	Δ Top3	Δ Top5	Δ Winner	Δ NDCG@3	Δ NDCG@5
OvertakeOpportunity	-0.0045	-0.0050	—	—	—	+0.0020	-0.0014
TrackTemp	-0.0008	-0.0005	—	—	—	-0.0024	-0.0023
QualiGapToPole	-0.0008	-0.0019	—	—	—	+0.0003	+0.0019
TeammateQualiGap	+0.0003	+0.0017	—	—	—	+0.0006	-0.0003
Quali_x_Overtake	+0.0007	+0.0010	—	—	—	+0.0018	+0.0021
TeamStrength	+0.0008	-0.0020	—	—	—	+0.0054	+0.0061
FP_LongRunVar	+0.0009	-0.0003	—	—	—	+0.0005	+0.0025
GridSpread	+0.0012	+0.0004	—	—	—	+0.0007	+0.0014
QualiPercentile	+0.0016	+0.0018	—	—	—	-0.0027	+0.0023
AirTemp	+0.0016	+0.0007	—	—	—	-0.0016	-0.0027
TeamDNFRate	+0.0016	+0.0009	—	—	—	-0.0004	+0.0023
DriverDNFRate	+0.0017	+0.0019	—	—	—	-0.0008	+0.0012
FP_LongRunPace	+0.0022	+0.0024	—	—	—	+0.0036	+0.0014
OvertakeIndex	+0.0023	+0.0016	—	—	—	-0.0050	-0.0013
Rainfall	+0.0029	+0.0032	—	—	—	-0.0024	-0.0024
Reliability_x_Rain	+0.0033	+0.0030	—	—	—	+0.0003	-0.0028
DriverForm	+0.0043	+0.0039	—	—	—	-0.0030	+0.0026
Driver_vs_Team	+0.0059	+0.0067	—	—	—	-0.0007	+0.0006

Table 11a. Leave-one-out feature analysis for XGBRanker v2. Delta = (model without feature) – (full model), averaged across 5 walk-forward folds. Sorted by Δ Spearman ascending (most damaging removals at top).

From this analysis, we see that **removing any individual features actually only results in minimal changes to model performance across all evaluation metrics**. This then suggests that **many features in the model are redundant**. Qualifying related variables is an example of this as *QualiPercentile*, *QualixOvertake*, and *OvertakeOpportunity*, are all highly correlated, thus **removing one doesn't significantly reduce predictive performance as the remaining features still capture similar information**.

The results indicate that the model is robust to the removal of individual features. Many predictors, as mentioned, provide overlapping information. Single-feature importance may be understated due to multicollinearity (multiple variables are correlated, hard for the model to isolate individual effects of each predictor). Even though individual features appear to have limited impact in isolation, this doesn't imply that the information that they represent is unimportant.

11.3 Variance Inflation Factor (VIF)

The full VIF analysis — what VIF is, why it is applied, the backward selection procedure, and a comparison of evaluation metrics before and after feature reduction — is presented in **Section 10.6** of this report. That section covers both the Plackett-Luce model (Section 10.6.2) and the XGBRanker (Section 10.6.3) in a unified treatment, including the removal sequence (*OvertakeOpportunity* → *DriverForm* → *Quali_x_Overtake*), the final 15 retained features, and the walk-forward evaluation metrics for each model before and after VIF reduction.

11.4 Overall

Overall, the leave one out analysis reveals that while the enriched feature set may contain many engineered predictors, **predictive performance is primarily driven by a small subset of core information**. The leave-one-out (LOO) analysis shows that **removing individual predictors results in only minor changes in performance**. This indicates a **high degree of redundancy** among features. This is also supported in the VIF analysis, which identifies significant multicollinearity.

The **VIF-based backward selection confirms that many predictors are strongly explained by others**, and their **removal actually does not meaningfully impact model performance**. Despite reducing multicollinearity, the VIF-reduced model still retains a large feature set, emphasizing **while features may be correlated, they still capture unique aspects of race dynamics**.

Across both the VIF and LOO analysis, we see **qualifying performance still being the dominant driver of predictive accuracy, with team strength and driver-related metrics providing additional but smaller contributions**. These are also consistent with the correlation analysis done prior, which showed strong relationships among engineered features. Overall, the results indicate that increasing feature complexity doesn't necessarily improve predictive performance. Carefully selecting features that provide independent information leads to more stable models without sacrificing accuracy.

12. Conclusion

This report has documented the full development trajectory of F1 race prediction models, from the original single-season Plackett-Luce model to the v2 Two-Stage XGBRanker and PL models with corrected methodology, evaluated using temporally valid walk-forward cross-validation across the 2022-2025 seasons.

12.1 Overall Model Performance

The table below summarizes mean walk-forward performance across all five folds for every model variant evaluated in this study. The **qualifying-position baseline** (score = $-\text{grid position}$) represents the simplest possible prediction: assume the race finishes in qualifying order. This is a competitive benchmark because grid position is the strongest single predictor of race outcome in Formula 1.

Model	Feat	Spearman	Kendall	Top3	Top5	Winner	NDCG@3	NDCG@5
XGBRanker (18 features)	18	0.6621	0.5240	0.6624	0.7686	0.5129	0.8909	0.8983
XGBRanker + VIF (15 feat)	15	0.6617	0.5189	0.6515	0.7805	0.4856	0.8901	0.9041
PL — Full (18 features)	18	0.6834	0.5456	0.6179	0.7661	0.4788	0.8879	0.9007
PL + VIF (15 features)	15	0.6847	0.5459	0.6205	0.7677	0.4621	0.8898	0.9023
PL + BIC (4 features)	4	0.6922	0.5554	0.6260	0.7644	0.4447	0.8873	0.9029
PL + AIC (5 features)	5	0.6944	0.5559	0.6407	0.7694	0.4356	0.8899	0.9017
Qualifying Baseline	1	0.6790	0.5404	0.6742	0.7727	0.6030	0.9001	0.9123

Table 12a. Mean walk-forward metrics across all 5 folds for all model variants.

12.2 Key Findings by Metric

Rank correlation (Spearman, Kendall): All model variants beat the baseline, and counter-intuitively, reducing features improves rank correlation. The AIC-selected 5-feature PL model achieves the best Spearman (0.694) and Kendall (0.556) of any model. This suggests the dropped features were adding noise to the global ordering rather than signal — a consequence of multicollinearity in the engineered feature set.

Top-3 and Top-5 accuracy: The qualifying baseline outperforms all models (Top3 = 0.674, Top5 = 0.773). Among actual models, XGBRanker is best on Top3 (0.662), followed by PL AIC (0.641). The rank-learning objective of XGBRanker, optimized directly for NDCG, is better suited for identifying podium finishers than the PL model's probabilistic ranking.

Winner accuracy: The baseline is the strongest predictor of the race winner (0.603) — the pole-sitter wins often enough in Formula 1 that no model consistently improves on it. XGBRanker (0.513) is the best among the trained models. All PL variants trail, with the more aggressively pruned BIC and AIC models performing worst (0.445, 0.436) as winner prediction benefits from the full feature set's granularity.

NDCG@3 and NDCG@5: The baseline again leads (0.900, 0.912), reflecting its strong positional accuracy. Model differences on NDCG are small (within 0.015) — all models capture the broad ranking structure similarly, with variation concentrated in the exact ordering of top finishers.

12.3 Model Selection Summary

The most significant methodological gains came from **walk-forward cross-validation** (eliminating temporal data leakage) and the **corrected QualiPercentile denominator**. Feature selection via VIF, BIC, and AIC consistently showed that the 18-feature set contains substantial redundancy — removing correlated features does not hurt and often helps rank correlation.

Qualifying position remains the dominant predictor of race outcome across all methods, consistent with the physical reality of Formula 1 where dirty-air effects limit overtaking. DriverForm and TeamStrength are the next most consistently selected features across all three PL selection methods (VIF, BIC, AIC), confirming that recent form and constructor competitiveness carry independent predictive signal beyond grid position alone.

The **2025 season** proved particularly challenging for both models due to a competitive reset — multiple teams became race-winning competitors, breaking the strong statistical patterns from the 2022-2024 Red Bull era. This is a regime-shift problem and highlights the fundamental limit of pre-race prediction: when the underlying competitive order changes substantially, historical training data cannot fully prepare the model.

Among all model variants, **no single model dominates on every metric**. For applications prioritising overall driver ordering (rankings, fantasy sports), the **PL + AIC model** is the strongest choice. For predicting top finishers and race winners, **XGBRanker** outperforms the PL variants and approaches the naive baseline. The qualifying baseline remains a surprisingly difficult benchmark to beat for exact positional accuracy, underscoring how much Formula 1 race results are determined before the lights go out.

12.4 Main Finding and Contribution

A recurring concern when evaluating these results is that the qualifying baseline is so competitive. It is important to recognise that this baseline is not weak — predicting that every race finishes in qualifying order requires no training, no features beyond grid position, and no model of any kind. The fact that it achieves 60.3% winner accuracy and Spearman 0.679 reflects a genuine property of Formula 1: **grid position is the single most powerful pre-race signal**, because aerodynamic dirty air makes overtaking physically difficult and dominant cars tend to qualify and race at the front. **Beating this baseline, even by a modest margin on held-out future seasons, is a meaningful result** — it means the model has extracted real predictive signal beyond what any analyst already knows from the starting grid.

The central finding of this study can be stated as follows: **a parsimonious 5-feature Plackett-Luce model, trained exclusively on past seasons, outperforms a qualifying-position baseline on rank-order accuracy for future races, while an 18-feature engineered model offers no improvement over this minimal set**. This has two implications. First, the predictable component of F1 race outcomes is captured by a small core of pre-race signals — qualifying position, team strength, circuit overtaking difficulty, driver form, and reliability. Second, additional engineered features beyond these five do not add predictive value; they introduce redundancy that, if anything, slightly hurts rank-ordering performance.

The difficulty of beating the baseline on **winner accuracy** is itself an informative finding rather than a failure. No pre-race model achieves winner accuracy above 52%, while the baseline achieves 60.3%. This gap directly quantifies **how much of a race winner's outcome is stochastic** — driven by safety cars, mechanical failures, pit-stop strategy, and on-track incidents that no model trained on pre-race data can anticipate. The remaining predictable component (correctly identifying likely winners roughly half the time from pre-race data alone) represents the ceiling of what systematic pre-race prediction can achieve in Formula 1 under current regulations.

Finally, the comparison between rolling and expanding training folds (Section 7.5) reveals that **more historical data does not uniformly help**. For 2024 predictions, adding 2022 data to the 2023 training set consistently improves both models, because the 2022-2023 seasons shared similar competitive dynamics. For 2025 predictions, the XGBRanker benefits from the full three-year training set while the PL model's winner accuracy declines — a consequence of the 2025 competitive reset where three years of Red Bull-era patterns became partially misleading. This suggests that in sports prediction, **recency and regime stability matter as much as sample size**, and that model-specific sensitivity to distributional shift should inform how training data is selected.